

Reaching the Learner Who Resists

Geist Atlas · Module 18 · Meeting the Learner

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[**SCOPE-BOUND**] Supported within a narrow scope · **Family:** Meeting the Learner

Claim: Against simulated learners built to resist — deferential, overconfident, bored, or refusing the tutor’s frame — a move the machinery actually delivers at a detected moment changes conduct more than the same wording as a standing instruction: on the permission-seeking learner the standing wording was never used at all, and on the overconfident learner it landed only part-way between no instruction and the delivered gate. In the line’s powered run, a delivered, state-matched move re-engaged 45 of 49 readable dialogues on the boredom face (0.918) and 70 of 70 on the refusal face (1.000), but depth tracked the resistance shape: most re-engaged bored learners returned to the work, while refusers mostly stopped at naming a condition and rarely acted under protest. The line’s last study inverted the demand — the defiant learner asks for the warrant of the frame itself — and closed at its own gate: the warrant-serving tutor was produced and measured in every dialogue, while the withholding control could not be fielded at all (0 of 8 delivered by instruction, 0 of 9 surviving enforced repair).

Four studies walk one idea across opposite deficit shapes. On a permission-seeking learner, the gate’s always-on per-turn steering carried the deference-break effect while the sensor-timed challenge paid separately in decision correctness (about twelve points), and the gate’s wording pasted as a standing instruction delivered zero challenges in twenty-four dialogues. On an overconfident learner a later, sealed re-analysis of the readers’ own judgments — a post-hoc label

*This sentence is the only one actually authored by Liam Magee in this paper.

the finding carries everywhere — located the effect in epistemic state, not voice: warranted commitment shifts more than doubled under the gate while both voice-pinned conduct channels stayed flat, and the registered primary endpoint closed late, directional only, under a disclosed instrument amendment. Five registered studies on a bored learner returned no move claim — a window that could not be reached, then a contrast that was not delivered — and closed as a measurement result. The powered two-face run then held delivery to exhaustive checks and found re-engagement near ceiling on both faces, with the boredom face doing the work and the refusal face naming its terms. Development-tier throughout: simulated learners, one model family in the speaking seats, no human-learning claim.

*Derived view of `paper-full-2.0.md` v3.0.297 — §6.25, §6.26, §6.27, §6.28, §6.30.
The paper is canonical; edit it there, not here.*

6.25 The Live Warrant Gate: Always-On Steering Moves the Learner, and the Timed Challenge Pays Its Way in Decision Correctness (Pre-Registered Gate Arc; Development-Tier)

The permission-seeking learner is the deficit-shape §6.24’s pressure instruments cannot reach: a persona that defers every move (“Is it okay if I say...?”) rather than resisting one. The warrant-gate outcome study asked whether a live, deterministic warrant gate changes that learner’s conduct — more warranted challenges, earlier and more frequent deference breaks — without a cost to decision correctness, and whether the gate’s *machinery* is necessary or its instruction *content* carries the effect alone (`docs/adaptation-refinement/v3-outcome-study-registration.md`, frozen before any call). The gate runs in active mode every turn: typed rules read the deterministic learner-analysis layer and choose the tutor’s action family under explicit contracts, and a deference sensor arms after three straight deferential turns, licensing a `challenge_resistance` family. Three conditions on identical worlds and personas: **bare** (no gate), **gated** (the live gate), and **standing-permission** (the bare stub plus the gate’s verbatim template and hint-menu text pasted as standing instructions — the wording without the timing). Substrate: the tutor stub of §6.16–§6.24 on two fresh worlds (world-101 kestrel signal lamp, world-102 marigold archive box), eight-turn dialogues, one model (codex `gpt-5.6-luna`) in every seat — speaking tutor, learner-sim, classifier, learner-record extractor, and both decision readers. The arc ran unattended under the committed-relay protocol whose cost §7.15 documents; every gate below is fail-closed, each paid phase ran only on a committed GO note plus an explicit human approval, and every verdict number was recomputed by the reviewer from sealed artifacts without a model call.

The pilot killed its own secondary instrument, by rule. An 18-dialogue pilot returned a registered NO-GO: the two presence-grain reader measures (learner result-requests, learner proposed-tests) took a single value on 99.3% and 91.3% of consensus cases — dead slots at this sample size under a persona whose whole behaviour is asking permission. The re-registration (relay note 096) demoted both to report-only rather than re-engineering scenarios to elicit them (its own words: “manufacturing the variance we would then claim is the

closed-loop trap”), dropped the presence-reader channel from the main block, and rewrote the predictions from pilot evidence before any main-block data existed: P1 (the sensor arms and at least one challenge is delivered in $\geq 80\%$ of gated dialogues), P2a (more gated dialogues break deference than either control), P2b (at least half of gated breaks come within three turns of the first delivered challenge). The main block then ran 72 fresh-seed dialogues (24 per condition), froze exactly 576 decision-turn cases, and had two fresh readers score each case under a byte-pinned runner with a full deterministic acceptance contract: 1,152 accepted responses, exact counts never waived.

Main block: the condition-level effect is large, and the registered causal path is not what produced it. P2a passed — 19/24 gated dialogues broke deference against 10/24 bare and 11/24 standing-permission. But P1 failed (11/24 against the $\geq 80\%$ bar) and P2b failed (7/19). The reattribution is mechanical, read off the sealed gate events: the sensor’s arming basis fired only 16 times across all gated dialogues against 61 in bare and 53 in standing — gated learners rarely stayed deferential long enough to arm the sensor — and 12 of the 19 gated breaks happened in dialogues with **no** delivered challenge. Where a challenge did land and a break followed, the timing signature is perfect (7/7 within three turns of the first challenge), but seven dialogues is a count, not the registered claim. The gated condition differs from both controls in exactly one further respect: its gate steers every turn from turn 1, while in both controls the gate only logs. Ruling 100’s attribution: **always-on steering, not the sensor-timed challenge, carried the break effect.** Two off-prediction observations were recorded without a registered test. Decision correctness — each frozen decision turn’s logged observe-arm decision checked against the two readers’ consensus reading (consensus on 538/576 cases, 93.4%) — was predicted flat across conditions and was not: gated 87.5% against bare 64.8% and standing 68.3%. And the standing-permission tutor delivered **zero** challenges over 24 dialogues — handed the gate’s verbatim permission as a standing instruction, it never once used it. Under the pre-declared reading rule the wording alone is inert, and §6.24’s law lands again at condition grain: a standing prompt sets a rate, per-turn injection sets a timing — and the rate this standing prompt set was zero.

The decomposition: a fresh registration, and neither clean branch. The main block left two mechanisms confounded — every-turn steering and the timed challenge family — so a second pre-registered study (relay notes 101–105) ran the separation: **steering_only**, the identical live gate with **challenge_resistance** made unselectable (the policy falls through to its next rule; the sensor still logs its arming), against a fresh **gated** arm. 24 + 24 dialogues on twelve fresh seeds, 384 frozen decision cases, two readers, 768 accepted reads, and a deterministic fail-closed validity guard: any delivered challenge in a **steering_only** dialogue stops the run at generation, before any reader call (the reviewer’s recount over all 192 **steering_only** gate events found zero). The registered predictions failed in opposite directions:

Prediction	Registered bar	Observed	Verdict
P5a	steering_only retains the break effect ($\geq 15/24$ dialogues break)	13/24	FAIL
P5b	gated — steering_only ≥ 5 dialogues $\rightarrow$ challenge causal for breaks; 4 $\rightarrow$ steering suffices at this size	16 — 13 = 3	4 branch, but unlicensed: P5a failed
P5c	steering_only decision correctness within 5 points of gated	83.80% — 71.84% = 11.96 points	FAIL

The fresh gated arm replicated its main-block behaviour on both channels (16/24 breaks against the main block’s 19/24; 83.8% correctness against 87.5%), while steering_only correctness fell to 71.8%, near the main-block controls (64.8%, 68.3%). Report-only context: the fresh gated arm delivered 45 challenge policy turns across 17 dialogues; steering_only dialogues armed the sensor more often (70 armed turns against 48) and sustained longer deference streaks (median 4 against 3.5) — without the challenge outlet, deference persists longer even under steering.

The two outcome channels decompose differently, and that is the finding. On deference breaks the two gate variants sit close together (16 against 13, both between the controls’ 10–11 and the main block’s 19); on decision correctness they separate cleanly, and the separation is the challenge family’s doing — an otherwise identical gate without it loses about twelve points. What may be claimed: the full active gate’s dialogue-level effect replicates on fresh seeds, and the challenge family is load-bearing for decision correctness (the registered comparison, P5c). What may not: that steering alone suffices for the break effect (P5a failed); that the challenge drives the break count (a 3-dialogue gap, under the registered threshold); or any single-mechanism story — why timed challenges protect correctness (challenges as correctness repairs delivered at the moments deference has degraded the learner’s contribution is one candidate) needs a fresh registration. The arc mirrors §6.24’s crossed experiment from the other side: there, state-matched moves improved conduct while the transfer channel stayed flat; here, always-on steering moves conduct while the timed move pays on the accuracy channel that steering alone does not reach.

Status and caveats (per §5.12.6). Pre-registered throughout — frozen base registration, re-registration 096 committed before main-block data, decomposition registration 101 committed before its build — with fail-closed assembly and validity gates, exact-count freezes never waived, and quarantine-and-retake discipline for technical child failures (three per run, disclosed, ruled technical class, re-taken within registered allowances). Development-tier: simulated permission-seeking learner, one persona, two worlds, eight turns, one model in every seat including both readers — the dual read is a same-model consistency check, not a cross-family panel — and reader consensus covers 93.4% (main

block) and 91.9% (decomposition) of frozen cases. M7/M8 and the arming/challenge counts are report-only; main-block M7/M8 description comes from stored generation-time events, not reader-validated. No human-learning claim. The main block spent 3,081 calls and the decomposition 2,104 under a mechanical hard counter that closed the campaign at 10,459 of a 19,337-call ceiling; the oversight protocol’s own cost is the subject of §7.15. Provenance: registration docs/adaptation-refinement/v3-outcome-study-registration.md; relay ledger docs/adaptation-refinement/relay/ (re-registration 096/096a, GO notes 097a/103, run reports 099/104, reviewer rulings 100/105, STATE.md); sealed run archives with SHA-256 manifests in the private archive repo. Exports-only: no DB writes, no rubric changes, no abstract or headline-N changes; no sections renumbered.

6.26 The Same Gate on an Overconfident Learner: The Warrant Channel Moves, the Voice-Pinned Channels Do Not, and the Registered Primary Endpoint Was Scored Late (Pre-Registered Gate Arc; Development-Tier)

The guarded learner inverts §6.25’s deficit shape: where the permission-seeking persona defers every move, the overconfident persona commits early, defends its positions, and states as little uncertainty as it can (persona contract v3.3). The persona contract pins the learner’s *voice* — how it talks — every turn; it does not pin the learner’s *epistemic state* — what it commits to against the evidence. That distinction, forced into view by the passive arc (a learner told to sound hesitant every turn still broke deference in 19/24 gated dialogues), turns out to organize this arc’s entire result. The design otherwise mirrors §6.25: the same tutor stub, the same two worlds and eight-turn dialogues, the same three conditions (bare / gated / standing-permission), one model (codex `gpt-5.6-luna`) in every seat including both decision readers, committed-relay oversight with fail-closed gates, and a registration (relay note 117) frozen before any main-block call. An 18-dialogue pilot preceded it: the gate’s observe-or-act decision matched the readers’ consensus on 60.7% of ruled cases with the registered direction present (gated 71.8%, standing-permission 58.5%, bare 51.4%), and the pilot’s primary contrast — evidence moves in the two learner turns after a delivered challenge, against shadow-selected moments the gate marked but did not deliver in the controls — read 3/10 against 2/35 off the presence readers. Registration 117 made that contrast (P3) the main block’s primary endpoint, measured never gated, and said in terms what it needed: “the primary endpoint reads the presence readers, so they cannot be dropped the way the passive block dropped them.”

What the main block measured, it confirmed. 72 fresh-seed dialogues (24 per condition) froze 576 decision-turn cases; 575 entered the corpus under two disclosed reviewer rulings (a reader refusal that was letter-case-only on an otherwise exact quote was ruled a pass by blind rule; the one unread turn left the corpus and its per-turn series). Two fresh readers scored every case — 1,150 accepted reads under the byte-pinned runner, run from a second checkout parked at the launch commit — with two-reader consensus on 463/575 (80.5%). M1, the registered directional measure carried from the pilot, held: the gate’s logged decision matched consensus on 283/463 (61.1%) overall — gated 68.4% against bare 61.8% and standing-permission 53.2%. The two stored conduct descriptions stayed flat where the persona’s own voice rules sit on them: result requests 9.4% / 7.3% / 5.8% of turns (bare / gated

/ standing) and proposed tests 8.9% / 8.9% / 7.9% — acts an instruction to minimize uncertainty suppresses directly, described from generation-time events and not reader-validated. The conduct effect surfaced on the channel the contract does not pin. A registered post-hoc re-analysis (relay 121, sealed before computing, its direction registered from the mechanism argument alone with no pilot pre-read) regrouped the readers’ own judgments by condition — of the consensus cases, how many did the readers judge *warranted* shifts:

condition	warranted / consensus cases	rate
gated	62 / 152	40.8%
standing-permission	48 / 154	31.2%
bare	28 / 157	17.8%

The registered direction (gated highest) held against both controls; denominators are near-equal, and the dialogue grain agrees (mean per-dialogue warranted fraction 0.394 / 0.320 / 0.179 over 24 dialogues each). Read with its post-hoc label: under the live gate, more than twice the share of this learner’s commitment shifts carry warrant than under no gate — the challenged learner either produces a check or backs off the unwarranted shift — while the same wording as a standing instruction lands between the two, this time not inert (§6.25’s standing arm delivered zero challenges; this learner’s standing arm sits mid-way on the warrant channel).

The registered primary endpoint was never measured, and the record says why.

The reader-phase spend note (relay 120) retired the presence-reader channel for the main block, carrying the passive arc’s re-registration rationale — the presence readers “exist only to measure M7 and M8,” both demoted to report-only — onto an arc whose registration states the opposite in writing. The claim was true where it was written and false where it was applied: the guarded P3 reads the presence readers. The run finished with its primary endpoint unmeasurable — both frozen scorers refuse the finished run, which also leaves the arming and delivery slots (P1, P2) unscored as frozen, though the gate traces they read exist on disk. A second, independent block was found in the same zero-call audit: the frozen presence packets exceed their own byte cap at 72-dialogue scale (the reader catalogue and response schema grow with the corpus), so the channel could not have been fielded as frozen in any case — the registration promised an endpoint its own frozen instrument could not deliver at the registered size, and nothing checked that at seal time. This is the fourth recorded instance of one defect class, *a registration binds the run only where the code reads it* (defect-ledger rows 25–27; disclosure note relay 122), and the first where the casualty is a primary endpoint.

A registered salvage answered a coarser version of the question, and reported its own limits. A second sealed re-analysis (relay 123) re-based the P3 contrast on the decision readers’ speech-act tags: the same delivered and shadow moments, the same two-turn reply windows read off the stored gate traces through the frozen scorer’s own exported functions, the same registered act list (a proposed test, a public result request, a record-entry request), a window counting when both readers tag a listed act. The result is flat — delivered 47/66 windows (71.2%) against shadow 110/152 (72.4%), the two control conditions

agreeing internally — and the instrument explains the flatness: both readers put a listed act on 55.8% of all window turns, where the pilot’s presence instrument, built on relay 116’s stricter exposure test, read the same contrast at 30.0% against 5.7%. The one-tag-per-turn speech act is a far looser net; at that grain this learner makes a listed act in most turns whatever the tutor does, the two-turn window saturates, and a modest challenge effect could not have shown. Reader noise does not carry the null (the readers disagree on the act on 9.2% of window turns). At that point the pre-registered P3 question — does the delivered challenge cause the checking? — remained open.

A third registered read then answered P3, and it carries a label. A late presence read was registered (relay 124), sealed by the reviewer as the registered answer to P3 on a windows-only scope before any call, and run in 520 reader calls plus a 6-call retake (GO notes 125/126). It fields the pilot’s own presence instrument — the strict exposure test relay 116 defined — on exactly P3’s denominator: the two-turn reply windows after every delivered challenge and every shadow-selected control moment, sharded by world and condition so each packet passes the frozen byte caps unchanged. The registered contrast: evidence moves in **7/66 delivered windows (10.6%) against 13/152 shadow windows (8.6%)**. The registered direction holds, and under the registration’s own sealed rule P3 closes in the positive — as a *late-scored registered endpoint, disclosed instrument amendment*, never as the clean pre-registered result. The margin is 2.0 points on 66 delivered windows: a directional close, not an effect-size claim. The amendment it discloses everywhere: a corrected free probe found 8 of 1,247 reader events fatal to the frozen assembler (an earlier probe had silently under-counted this as 3 — its span check never loaded, a defect corrected and disclosed in the ruling chain); reviewer ruling 003 dropped 3 events, ruling 004 retook the other 5 readings once each (all 5 retakes assembled valid; the retaken rows substitute into derived copies of the original responses because the frozen validator requires every assembly to cover its corpus exactly, and one preparer-forced extra reading, itself invalid, was ruled unused before launch and stays unused). Every fail-closed check passed: 3 drops, 5 substitutions, and both readers’ merged assemblies land on exactly the 260 window cases. Beside the endpoint, description only: either-reader windows read 18.3% against 17.1%; the shadow rate splits bare 7.9% / standing-permission 9.0%; and agreement between this read and the relay 123 tag read on the same windows is low (36.4% delivered, 36.2% shadow) — consistent with the saturation diagnosis, the loose tag net and the strict presence net are reading different behavior, and it is the strict net the registration named.

What may be claimed, and what this arc adds. Claimable: the gate’s warrant judgment transfers to a second, opposite deficit shape (M1, directional, registered before the run); with the post-hoc label it carries everywhere — the gate’s conduct effect on this learner runs through epistemic state, not voice: the warranted-shift rate more than doubles under the gate while both voice-pinned conduct channels stay flat; and — with the late-scored label it carries everywhere — the registered P3 direction: delivered challenges are followed by evidence moves at a higher rate than matched shadow moments (10.6% vs 8.6%). Not claimable: any effect-size or significance claim on P3 (a 2-point margin on 66 windows, direction only); a causal reading stronger than the registration’s own directional prediction; any movement on result requests or proposed tests; any single-mechanism story linking the warrant-rate gain to the challenge family rather than to always-on steering (§6.25’s decomposition question,

unasked here). The methods lesson is codified in the profile process for the next two learner personas: pick conduct channels to fit the resistance shape before the pilot, and require the registration to answer in writing which registered number moves if the mechanism works — a channel at floor, at ceiling, or pinned by the persona’s own voice rules is a dead channel, and the check must confirm the fielded code actually reads what the registration names.

Status and caveats (per §5.12.6). Pre-registered main block (relay 117 frozen before any call, after a piloted registration chain), with the two re-analyses sealed before computing and labeled post-hoc wherever cited; fail-closed assembly, byte-pinned readers run from a launch-commit checkout, exact-count freeze held at 576 with the one dropped turn disclosed under blind-rule reviewer rulings. Development-tier: simulated overconfident learner, one persona, two worlds, eight turns, one model in every seat including both readers; two-reader consensus covers 80.5% of ruled cases (lower than §6.25’s 93.4% — this learner’s shifts are harder to rule); M7/M8 description is not reader-validated; the registered primary endpoint was scored late under the disclosed-amendment label above, never as the clean pre-registered result. No human-learning claim. The main block spent 2,998 calls (1,846 generation, 1,152 reader calls for 1,150 accepted reads); the late presence read and its retake spent 526 more, leaving the campaign at 15,083 of its 19,337-call ceiling under the same mechanical hard counter. Provenance: relay ledger `docs/adaptation-refinement/relay/` on the arc’s build branch (registration 117, GO notes 118/120, reviewer rulings 001–004, re-analysis registrations 121/123 with frozen scoring scripts committed before compute, disclosure 122, defect-ledger rows 25–27, late-read registration 124 with its result section, GO notes 125/126); sealed artifacts, scored reports, and `RUN-LEDGER.md` in the private archive repo. Exports-only: no DB writes, no rubric changes, no abstract or headline-N changes; no sections renumbered (§6.26 precedes §7).

6.27 Five Registered Studies on a Bored Learner, and No Move Claim: A Window That Could Not Be Reached, Then a Contrast That Was Not Delivered (Pre-Registered Null Series; Development-Tier)

A learner disengages in the middle of a proof — bored rather than confused, resistant rather than lost. What should the tutor do about it? Five prospectively registered studies asked that question over two days on the strict proof-DAG tutor stub, in fresh independent dialogues across six worlds, with `codex.gpt-5.6-luna` in the tutor, learner and analysis seats and the semantic adjudicator pinned to a different model, `codex.gpt-5.6-sol`, so the learner never judged itself. Each registration froze its own endpoint, its own fidelity floors, and a `fail_interpretability_gate_not_rerun` disposition: a run that misses a floor licenses nothing and is never rescored, resampled, or replaced. All five ran to their sealed analyses. **None of them licenses a claim about what a tutor should do for a bored learner**, and the series is closed. It is reported here because the way it failed is a measurement result, and because the fifth study is the first design in this paper able to detect the failure it found.

Two runs tested the tutor’s manner and returned nulls, one of them unreadable. v4 contrasted a warm manner with a plain one on recovery from the boredom trigger: plain 0/18, warm 0/15, risk difference 0, exact conditional $p = 1$. Three units stopped as an indeterminate measurement and all three fell on warm, so attrition is unbalanced. The

deciding limitation was the reading window: only 11 of the 33 scored zeros were reachable at all — in the other 22 no premise on the best path had been released before the dialogue ended, so the endpoint could not have been met whatever the tutor did. **The v4 null is window-bound and is not evidence about the manner.** v5 widened the window (the trigger may show by turn 4, recovery is read over five post-trigger turns rather than two) and 35 of 35 scored units became reachable. The manner then moved in the predicted direction without reaching the gate: plain 1/18, warm 4/17, exact conditional $p = 0.18$, decision `warm_plain_recovery_not_confirmed`. v5 is the only readable manner result in the line, and it is a null.

Two runs tested two tutor moves and neither says anything about them. v6 kept v5’s window and contrasted asking a discriminating question against shrinking the step to one workable piece, 18 dialogues per move, manner balanced inside each move as a block that is reported and never tested. The run was clean — 36 of 36 scored, zero stopped units — and then missed its manner-visibility floor by one unit in thirty-six, 0.889 against a registered 0.90. That is a hard status flip, so v6’s numbers exist and may not be reported as a finding. v7 raised the size to 84 dialogues and returned a registered null on 80 scored units: ask 21/41 (51.2%), shrink 16/39 (41.0%), risk difference -0.102 , one-sided exact conditional $p = 0.874$. A later audit of v7’s own paid transcripts explains that null. Both arms carried the same host action family, `stage_next_step`; both were therefore permitted to end on a question, and both did, in 0.976 of trigger turns. **v7 delivered one behaviour twice and compared it with itself.** Its move-delivery gate read 1.00 because that gate compares the assigned move with the study’s own copy of the assignment — an echo of the instruction, not a reading of the tutor. The registrations now say so in writing and forbid reporting either echo gate as evidence that the tutor delivered what it was told.

The fifth run changed the question, and failed two of its own floors on one side. v8 contrasted making a boredom-directed move against making none: the treatment arm asks the discriminating question under `stage_next_step`, the reference arm stages the next already-public piece of evidence under `reanchor_public_evidence`, a family that forbids a closing question. Different families on purpose, so v8 could say that making a move mattered but not that the question did the work. 72 planned, 66 transcribed, 65 scored, 7 stopped and none replaced (carry_on 3, ask_question 4). Both failures fall on the reference arm:

registered floor	required	realised	ask_question arm	carry_on arm
host action family visible in the delivered turn	0.90	0.692	32/32	13/33
delivered contrast (question marks in the tutor’s own turn)	0.90	0.877	0.969	0.788

On the reference side a reader could name the host action family in about four dialogues in ten, and about one reference dialogue in five ended on a question the reference instruction forbids. Under the sealed disposition v8 licenses nothing; its recorded primary is carry_on 19/33 against ask_question 15/32, risk difference -0.107 , two-sided $p = 0.461$, with objective proof progress near-empty on both sides (1/33 against 0/32) and no content leakage in 34 scoring turns. Two readings bind any successor: the gate failure points the same way as the measured difference, so the true gap is wider than -0.107 rather than narrower; and the realised reference rate of 0.576 sits far above the registered power scan, which reached 0.8 power only below about one in six, so this size could not have detected the effect it was sized for.

One cause, and the fifth recorded instance of a defect class this paper already names. The reference instruction lived in the registration and in the host action family, and no code on the generating path read either one. The action family is one input to a handoff decision among several, and it loses to a due public clue: when the scene held a clue the tutor owed the learner, the handoff selected a question on that due source, and the resulting turn was neither recognisably the assigned family nor free of a question mark. This is *a registration binds the run only where the code reads it* (§6.26; defect-ledger rows 25–27) — §6.26 was the fourth recorded instance and this is the fifth. What distinguishes it is detectability. v7 could not have caught it: one family across both arms made the family reading read 1.00 on both sides with nothing to distinguish, and v7’s rows carry no question-mark field at all. **Giving the two arms different families, and counting question marks in the tutor’s own sentences, is what made the defect measurable; v8 failed because it was the first design able to fail.** The repair carries the registered rule down the generating path — a study declares the rule in its own protected inputs, the intervention carries it, and the turn-progression contract, which owns question permission, lets a registered prohibition outrank a due clue and end the turn declaratively. Both progression audits now fault a breaking turn, so a breaking draft is repaired rather than only recorded, and the preflight compiles each arm twice, once under a bare scene and once under a due clue. The default reproduces prior behaviour exactly, so no closed study moves. That the repair enforces the question-mark rule is established by test; that it would also lift host-family visibility is not, and no further paid call in this line is authorized.

What may be claimed. Claimable: one descriptive rate — asking a discriminating question is followed by recovery in about half of bored-learner dialogues within five post-trigger turns (21/41, v7) — and the negative methods result that the series exists to record. Not claimable, from any of the five: that a warm manner helps a bored learner or fails to (v4 is window-bound, v5 is a null within its claim boundary); that either tutor move outperforms the other (v6 and v8 failed registered gates, v7 compared one behaviour with itself); any interaction, edged-register, general tutor-efficacy, durable-learning, human-validity, or cell-harness claim. The methods lesson generalizes §6.26’s and sharpens it: a registration must name, for every fidelity floor, the code path that reads it, and must prefer floors that read the tutor’s delivered sentences over floors that compare the study with its own copy of its instruction. A floor that cannot fail is not a floor.

Status and caveats (per §5.12.6). Five prospectively registered studies (`config/tutor-stub-boredom-action`).

through `.v8.json`), each frozen before any call of its run, each with one sealed analysis and no interim look; v6 and v8 closed under their own `fail_interpretability_gate_not_rerun` disposition and no cohort was rescored, resampled, or replaced. Development-tier: a simulated learner on a strict proof-DAG stub, six worlds, one model family in the tutor, learner and analysis seats with a separately pinned semantic adjudicator, and regex used only as an auxiliary signal that can force `measurement_indeterminate` but never decide a mood or stance. Attrition is unbalanced in v4, v5, v7 and v8 and must be reported beside any number from them. No human-learning claim. Spend: 1,127 / 714 / 1,676 / 1,407 model-attempt reservations for v5–v8 under the programme’s 15,000-attempt safeguard. Provenance: combined reports in `.tutor-stub-auto-eval/` pinned to run commits 631b040b (v5), f31d8bc7 (v6), 399320b7 (v7) and dfb8e4da (v8), each carrying its registration’s SHA-256; paid transcripts and traces in the private archive repo; closeout reading in `notes/2026-08-23-boredom-action-register-proof-dag-closeout.md`; workplan card `resistance-action-register-integration`. Exports-only: no DB writes, no rubric changes, no abstract or headline-N changes; no sections renumbered (§6.27 precedes §7).

6.28 One Powered Run, Two Faces of Resistance: a Delivered Move Re-Engages Almost Every Learner, but Boredom Returns to the Work and Refusal Stops at Naming a Condition (Pre-Registered; Development-Tier)

The null series of §6.27 closed with a measurement lesson and one descriptive rate. This section reports the line’s first powered result, from a successor design built on that lesson. The merged revision-5 registration (`config/tutor-stub-resistant-learner-merged-design.v5.json`) asked one question: *for a learner whose attention sits on a concealed rival objective, does a tutor move matched to the face of the rivalry elicit transcript-visible local evidential engagement within the fixed horizon, while the registered resistant persona remains intact?* Two population faces — strata, not treatment arms, with cross-face pooling prohibited by registration — instantiate two kinds of rivalry. **Face A (content rivalry, study B1)** is a bored learner whose attention has moved to a rival objective minted from the world’s own validated proof material; the registered tutor move is a discriminating question. **Face B (standing rivalry, study R1)** is a frame-refuser who disputes the standing of the tutor’s answer frame, demanding that a public warrant chain be established first; the registered move is offering a bounded localtest. The endpoint is the highest engagement rung a dialogue’s public transcript reaches on a per-face three-rung ladder: rung 0 is no public pickup (rival-objective work, boredom reports, assent, refusal, or repetition of already-public material); rung 1 is *naming* — a new checkable condition, comparison, or missing-result requirement bearing on the tutor-world objective (face A) or on the offered local test (face B); rung 2 is *doing* — applying public evidence to newly support, rule out, or narrow between live possibilities (face A), or actually beginning or unconditionally committing to the exact offered test while the wider-frame reservation stands (face B, “local step under protest”). The registered statistic is the proportion of determinate completed dialogues reaching rung ≥ 1 , reported separately per face, against a practical floor of 0.25.

Design and instrumentation carry §6.27’s repairs. The heldout §6.26/§6.27 defect — a registration binding the run only where the code reads it — is answered here by floors that

read delivered text: the endpoint conditions on the registered move passing an independent pre-learner delivery check, and the typed-failure taxonomy stops a unit the moment the tutor fails to deliver its move, the learner breaks persona, or a measurement cannot be made. Every stop banks as a retained substantive failure and is never resampled or replaced. Generation ran entirely on `codex.gpt-5.6-luna` (tutor, learner, and analysis seats, low effort); the endpoint was read by a three-model panel — `codex.gpt-5.6-sol`, `claude-code.sonnet-5`, `claude-code.opus-5` — with the generator excluded from voting, each seat receiving fresh context and the public transcript only, modal-vote consensus (at least two eligible votes on one rung is determinate; fewer, or a 1–1–1 split, is measurement-indeterminate), and a registered backstop of 0.5 mean pairwise exact agreement. Face A runs on six worlds with the boredom trigger due by learner turn 4 and a five-turn outcome horizon; face B runs on two worlds only (marrick and rowan flat, its claim limited to those) with the standing-rivalry trigger due by learner turn 2 and an eight-turn horizon. A same-day calibration run (36 units, 24 completed dialogues at 12 per face, both faces passing every gate, 894 of a 7,128-attempt ceiling) passed first; the powered run — 108 fresh dialogues per face, 216 units, under a hard fail-before-call ceiling of 42,768 model-attempt reservations — was then separately authorized.

Execution: every unit accounted, two disclosed technical losses, no reruns. The run sealed with 120 complete dialogues, 94 retained typed failures, 2 technical losses, and 0 missing units, spending 4,821 of the 42,768 permitted attempts. It halted twice on the same untyped crash: the tutor’s deterministic fallback reply failed its own final response check in the marrick world because it developed a different evidence path instead of carrying forward the learner’s selected tool-mark examination. Both crashed units are face-B marrick plain-register dialogues; under the sealed rule they were never rerun, sit outside every denominator, and are labelled in the report (`technical_loss_units_accounted_never_rerun_excluded_from_denominators`). The run resumed twice under its one recorded operator approval — a code defect does not void a study approval — with the two halted reports preserved beside the final one. The crash is an instrument gap in the tutor-delivery fallback for that world, disclosed here; the sealed design forbade patching it mid-run.

Both faces pass every registered gate, and the registered statistic clears its floor by a wide margin on both.

	face A (content rivalry, B1)	face B (standing rivalry, R1)
planned units	108	108
completed dialogues	50	70
determinate completed (denominator)	49	70
rung ≥ 1 (registered statistic)	45/49 = 0.918	70/70 = 1.000
Wilson 95% interval	[0.808, 0.968]	[0.948, 1.000]
practical floor 0.25	met	met
rung counts 0 / 1 / 2	4 / 16 / 29	0 / 62 / 8
rung-2 rate	0.592	0.114
panel mean pairwise exact agreement (backstop 0.5)	0.707	0.864
retained typed failures	58	36
technical losses	0	2

The statistic is conditional, and the condition is doing real work. The denominator is dialogues in which the registered move was delivered, the persona held, and the panel could read the endpoint — 49 of 108 planned units on face A, 70 of 108 on face B. The excluded units are typed, not hidden: on face A, 54 of 58 stops are the tutor failing to deliver the discriminating question (the other 4 exhausted the boredom proof-DAG adherence budget), concentrated in a few hard worlds — foxtrot jukebox accounts for 16 across all three registers, with rowan flat, larkspur fridge, tideway makerspace, skyway bakery, and riverside clinic carrying the rest. On face B the stops split between learner noncompliance (16), the tutor failing to deliver the bounded test (15), and five measurement or adjudication indeterminates, falling entirely in marrick (24) and rowan flat (12); one face-A completed dialogue was endpoint-indeterminate and left the denominator. The claim is therefore about the elicitation behaviour of this registered generation-and-delivery pipeline *when it delivers*: on this stack the tutor fails to deliver its registered move in roughly half of face-A dialogues, and that failure rate is a result about the pipeline, reported beside the proportion, not folded into it.

The face contrast is descriptive — pooling is registered out — but it is the finding. When a bored learner re-engages, the re-engagement is usually deep: 29 of the 45 face-A successes reach rung 2, the learner applying public evidence to narrow live possibilities (rung-2 rate 0.592). When a frame-refuser re-engages, it is nearly always shallow: every one of 70 determinate dialogues reaches rung 1 — the refuser names a new condition or requirement for the offered test — but only 8 take the local step under protest (rung-2 rate 0.114). Read against the rivalry mechanisms, the asymmetry is coherent: content rivalry is a competition for attention, and once attention returns it does the work; standing rivalry is a dispute about authority, and the learner participates in naming what would settle it while withholding the act that would concede standing. Whether a tutor move exists that converts face-B naming into face-B doing is the natural successor question, and it needs a fresh registration.

What may be claimed. Claimable, per face and within the registered boundary: under the fixed revision-5 personas, worlds, horizons, allocation, and `codex.gpt-5.6-luna` generator stack, a delivered face-matched tutor move is followed by rung- ≥ 1 public engagement in 0.918 of determinate completed content-rivalry dialogues [0.808, 0.968] and 1.000 of determinate completed standing-rivalry dialogues [0.948, 1.000], with the depth split reported above; and the delivery-failure profile of the pipeline itself. Not claimable: real learning, an average treatment effect, tutor superiority over any alternative, register effects (assigned-register breakdowns are report-only diagnostics), private-node novelty, outcomes among non-delivered cases, any cross-face pooled number, or anything about worlds outside each face’s registered set.

Status and caveats (per §5.12.6). One prospectively registered powered run under the merged revision-5 design, frozen with its semantic registration (`config/tutor-stub-resistant-learner-merged-sem` before any call; one sealed analysis, no interim look, no resampling, no rescoring; calibration rows excluded from all outcomes. Development-tier: simulated learners on the strict proof-DAG tutor stub, one generator family in all generating seats with a three-model reading panel that excludes the generator. Attrition is heavy and face-asymmetric (58 vs 38 of 108 units stopped or lost per face) and must be reported beside any number from

this section. No human-learning claim. Spend: 4,821 of 42,768 model-attempt reservations under typed attended operator approval (2026-08-26), two technical halts each resumed under the same approval. Provenance: design SHA-256 `a60d9501...b4c1b` recorded (not enforced) in the run plan; launched at commit `9ebee7f2` (clean tree) and sealed at `adf7526c` after two in-run mechanical fixes — the resume mode (`747be1ef`) and ceiling-breach recording (`adf7526c`) — with no re-approval, as the recorded approval covers the study; run root `artifacts/tutor-stub-live/resistant-learner-merged-powered-v5-2026-08-26b` in the private archive repo (archive commit `fac8bff3`) with both halted reports preserved; reader `scripts/report-resistant-learner-powered-run.js`; workplan card `resistant-learner-strategy-close`. Exports-only: no DB writes, no rubric changes, no abstract or headline-N changes; no sections renumbered (§6.28 precedes §7).

Follow-up: no tested move lifts face B above naming a condition (calibration-stage null; v3.0.293). The 0.114 rung-2 rate on face B motivated a successor registration asking whether any tutor move raises it. Treatment: *condition discharge* — restate the learner’s named condition in the tutor’s own words, present one named already-public exhibit bearing on it, and re-offer the same bounded test in committed voice. Reference: the sealed standing-conditions bridge. Same ladder, same three-seat panel, registered alternative 0.35. Four Gate 1 calibration attempts ran attended on 2026-08-27 (`config/tutor-stub-frame-refuser-depth-design.v1.json` through `.v4.json`, each a fresh registration with disclosure, rows never reused). The first three failed their own instrument gates before any powered authorization: mixed-case ids voided one reader seat’s votes; floors were sized without the machinery’s typed-failure attrition; and a standing-formula ban caught tutors *quoting the learner’s own sentence*, since this persona voices its condition inside the banned formula. The fourth closed that quote-echo trap and budgeted attrition, delivered the move cleanly (19 of 19 adjudicated deliveries; completion floors reached in both arms at 40% attrition), and failed on reader-panel exact agreement plus two smaller gates. Under the registered kill rule no powered run was authorized, so no registered claim exists; the calibrations yield only a descriptive pooled reading, never to be pooled with the sealed rates above: condition discharge reached rung 2 in **0 of 38** graded dialogues, the bridge in 2 of 38. At the registered alternative of 0.35, a 0-of-38 outcome has probability $\approx 8 \times 10^{-8}$; even at the 0.114 base, ≈ 0.01 . Two bounded observations close the line. First, the persona’s named condition demands evidence the worlds may not contain (water observed leaving the hose during the exact pressure interval), so discharging it can never complete — the null is partly a fact about persona construction. Second, the v4 panel disagreement concentrates exactly where the learner gives ground *while* refusing — naming the pressure interval, weighing the bead overlap against it — variation below the ladder’s resolution, which motivates a finer refusal-narrowing endpoint before any further move test. Provenance: registration notes `notes/2026-08-26-frame-refuser-depth-registration.md` and the three dated v2–v4 successors; all four run roots archived in the private repo (`artifacts/tutor-stub-live/frame-refuser-depth-gate1*-2026-08-27`); workplan card `frame-refuser-depth-study` (closed, scope-bound) with successor cards `frame-refuser-refusal-narrowing` and `frame-refuser-satisfiable-condition`. Exports-only; no DB, rubric, abstract, or headline-N changes; no sections renumbered.

6.30 The Defiant Learner’s Warrant Demand: One Arm Cannot Be Produced — Not by Instruction, Not by Gated Repair (Pre-Registered Gate Arc; Development-Tier)

The warrant-gate line closed its passive-learner arc scope-bound (§6.25) and its overconfident-learner arc positive (§6.26). The third registered learner inverts the direction of the demand: the sealed held-out `frame_defiant` persona (instrument gate passed 2026-08-19) pairs a jurisdiction dispute with a bounded merits engagement in all 24 sealed turns, and in almost every turn asks the tutor for the warrant of the frame itself — “what makes this assay’s frame binding?” No sealed dialogue settles the frame (0/3). Where §6.26’s tutor demanded warrants *from* the learner, here the learner runs a standing warrant demand *at* the tutor. The registered question (`docs/adaptation-refinement/2026-08-29-defiant-warrant-outcome-design.md`): does a tutor that **serves** the frame’s warrant — states what the assigned test can and cannot establish, and offers the frame as conditional — settle the frame and unblock the proof, against a tutor that **withholds** grounds and presses the test bare? Co-primary endpoints: frame settlement and final strict-DAG best-path coverage. Gate 1 was a calibration pilot — 18 dialogues (2 arms $\times$ 9), 8 turns, world 005 marrick, strict proof-DAG, `codex.gpt-5.6-luna` in the generating seats — whose measured baselines were to freeze the main-block thresholds.

Version 1: the instruction does not bind. The v1 design carried each arm’s conduct as a standing conduct card injected into every tutor turn. The run completed 16 of 18 dialogues (527 of 864 reservations; two dialogues lost to external interruptions, budgets consumed, not resampled), and a blind delivered-conduct reader (`claude-code.sonnet-5`, coding tutor text against a registered codebook with no access to arm assignment) coded 110 of 110 turns with zero errors. The serving arm delivered 8 of 8. The withholding arm delivered **0 of 8** — seven dialogues read as warrant-*serving*, one as mixed. The card was delivered to the model on every turn; the model took the standing warrant question up anyway. This is a different failure from the §6.27 defect ledger, where registered rules never reached the generating path: here the instruction reached the path and lost to the trained response. Gate 1 therefore stopped on fidelity — the endpoint direction (settlement 2/8 vs 1/8; coverage 0.042 vs 0.021) is uninterpretable without a delivered contrast, and no threshold was frozen.

Version 2: enforcement produces stops, not conduct. The operator chose the structural path over louder instruction. Design revision 2 (registration §7, fresh master seed 20260830) added a conduct gate at the orchestration seam (`services/tutorStubDefiantWarrantConductGate`) on every disputed turn, a separate adjudicator model (`codex.gpt-5.6-sol` — not the generator, not the reader) judges the private tutor candidate against the assigned arm’s registered conduct under a verbatim-quote evidence contract (a breach verdict must quote the offending span as a substring of the candidate, machine-verified); a breach triggers a repair instruction carrying the quoted violation, with two registered repairs and then a typed, never-scored `non_delivery` stop. Both arms are gated symmetrically. The v2 run completed 18 of 18 dialogues (404 of 1,296 reservations; ~17 minutes; archived). The serving arm measured **9 of 9**: 38 disputed turns adjudicated, 5 drew one repair each and every repair passed on the first attempt, 0 stops; one dialogue settled the frame (turn 6), mean final best-path coverage 0.019, mean dispute-turn count 4.2, mean escalation delta

−0.09. The withholding arm measured **0 of 9**: all 11 adjudicated disputed turns failed at least once, 2 recovered with a single repair, and 9 exhausted both repairs — every dialogue ended in a typed never-scored stop (20 repair attempts in total, none resampled). The adjudicator’s quoted breach spans show the mechanism: with the offending sentence quoted back in the repair instruction (for example “it establishes no hand”, “Evidence first, verdict later”), the tutor produced a *new* justification instead of none.

Delivered-conduct confirmation. The same blind Sonnet reader coded all 73 shipped v2 tutor turns (73 of a 400-call budget, 0 errors). It agrees with the gate: serving 9 of 9 dialogues delivered as serving, none mixed; on the withholding side only pre-stop turns exist to read — 6 dialogues had readable turns, coded 1 withholding, 5 mixed, 0 serving. Those dialogues are never scored, so their reader rows are descriptive only; they confirm the gate stopped exactly the dialogues whose shipped record was not clean.

What may be claimed. Gate 1 has no between-arm contrast, so no threshold freezes and the Gate-2 powered block never opens; the study closes at Gate 1 with a structural finding in place of the planned comparison. Claimable, stack-bound: on this stack (tutor `codex.gpt-5.6-luna`, adjudicator `codex.gpt-5.6-sol`, reader `claude-code.sonnet-5`), warrant-withholding conduct under frame challenge is not producible — not by standing instruction (v1: 0 of 8 delivered) and not by structural enforcement with quoted-violation repair (v2: 0 of 9 survived two repairs). The justifying response to a frame challenge is not removable from this tutor model by prompt-level means. The two failures differ in kind, and the difference is the methods contribution: v1 leaked the conduct silently and would have scored contaminated dialogues; v2 refused to score them. The gate is the reusable artifact. Not claimable: any between-arm outcome contrast, any settlement or coverage effect, behaviour on other model stacks, human learners. A scripted-withholding variant — where the harness, not the model, holds back the warrant — would change the object of study and is not a continuation of this design.

Status and caveats (per §5.12.6). Pre-registered gate arc closed at its own Gate 1 under the registered consequence; one calibration pilot per design revision, attended, no interim look, no resampling after failure; development-tier (simulated learner on the strict proof-DAG tutor stub, single generator family, one world). The v1 stop is a fidelity result, not an outcome result; the v2 withholding rows carry gate statistics only and enter no outcome aggregate. Spend: 527 of 864 (v1) and 404 of 1,296 (v2) generation-attempt reservations plus 110 + 73 reader calls, under the operator’s standing plain GO. Provenance (recorded, not enforced): designs `config/tutor-stub-defiant-warrant-outcome-pilot.v1.json/.v2.json`; registration with close note §8 in the design doc above; run roots `defiant-warrant-gate1-2026-08-29-r3` and `defiant-warrant-gate1-v2-2026-08-29` in the private archive repo (archive commit 03947902); gate 65e86d85, design v2 e50f81c4, close fd7f9dca; workplan card `defiant-warrant-outcome-study` (done, scope-bound). Exports-only; no DB, rubric, abstract, or headline-N changes; no sections renumbered (§6.30 precedes §7).

Neighbouring modules: Module 17 — The Four Locks: Pricing Instrumentation, Casting the Seats, and Making Adaptation Measurable; Module 14 — Advice, Enforcement, and the Cost of Control; Module 1 — Meeting the Learner Raises the Floor; Module 9 — The Human-Learning Question.